

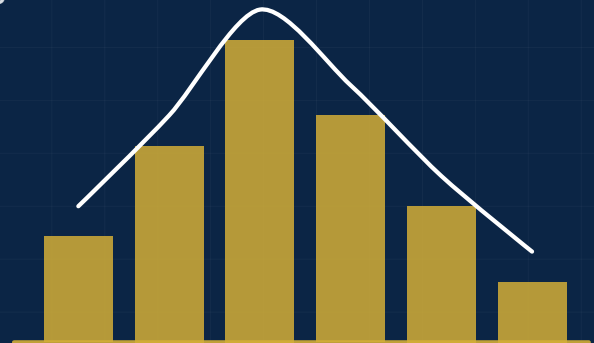


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CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 5

Venn & Tree Diagrams

Venn: fill from the middle outwards, then read off. Tree: write every branch (pairs add to 1), multiply along, add between routes.



5

VARIANTS

24

PAST-PAPER QUESTIONS

56

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Two-circle Venn

V2 Three-circle Venn

V3 Trees: independent

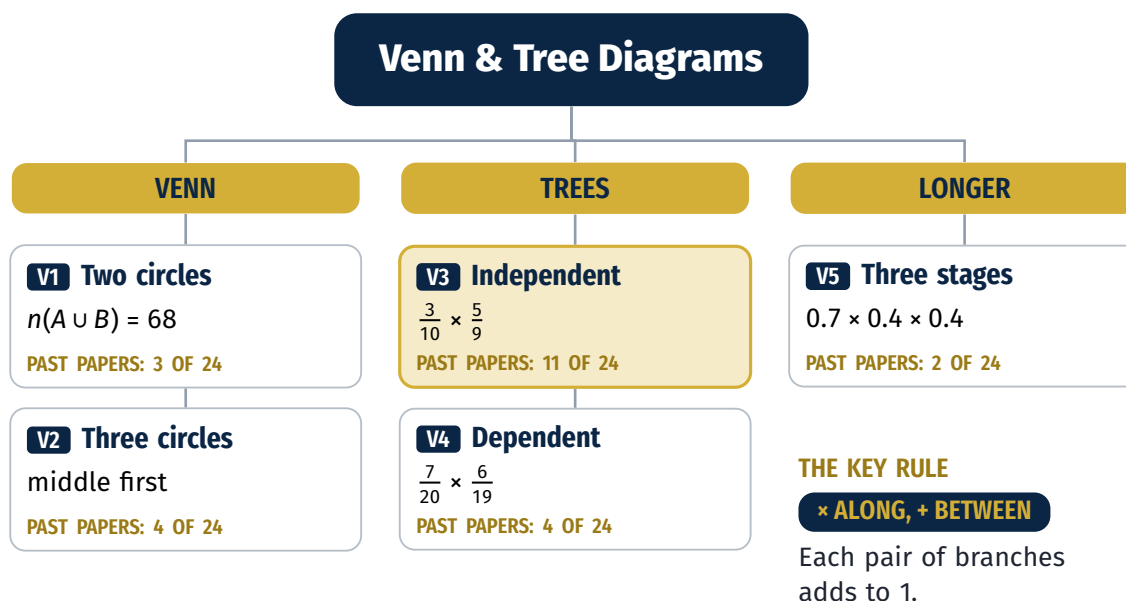
V4 Trees: dependent

V5 Longer trees

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Venn & tree diagrams at a glance



Your notes


How often does each variant come up?

24 past-paper questions (Jan 2020 → Nov 2025); 56 practice parts — grouped by type.

	PAST PAPERS (OUT OF 24)	PRACTICE PARTS	
V1 Two-circle Venn	3	18	
V2 Three-circle Venn	4	4	
V3 Trees: independent	11	14	
V4 Trees: dependent	4	8	
V5 Longer trees	2	4	

WHAT THIS TELLS YOU A tree diagram is on almost every paper (Q12–15, 4–5 marks): complete it, then one probability. A three-circle Venn diagram turns up about once a year as a bigger question.

BEFORE YOU START

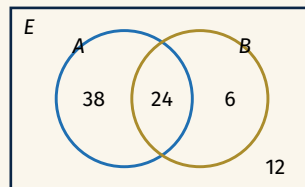
The big idea



Pictures that count for you.

A **Venn diagram** puts each item in exactly one region — fill the overlap **first**.

A **tree diagram** shows events in order: **multiply** along a route, **add** the routes you want.



Set words

$A \cap B$
and
In both (overlap).

$A \cup B$
or
In either (both circles).

A'
not A
Outside A.

Given A
 $\frac{\cdot}{n(A)}$
Only A is the total.

Tree rules

Branch pairs
 $p, 1 - p$
Each pair adds to 1.

Along
 $\times$
One then the other.

Between
 $+$
Different routes.



CALCULATOR FIRST, THEN CHECK

Check: all the end-of-route probabilities add to 1; all Venn regions add to the total.

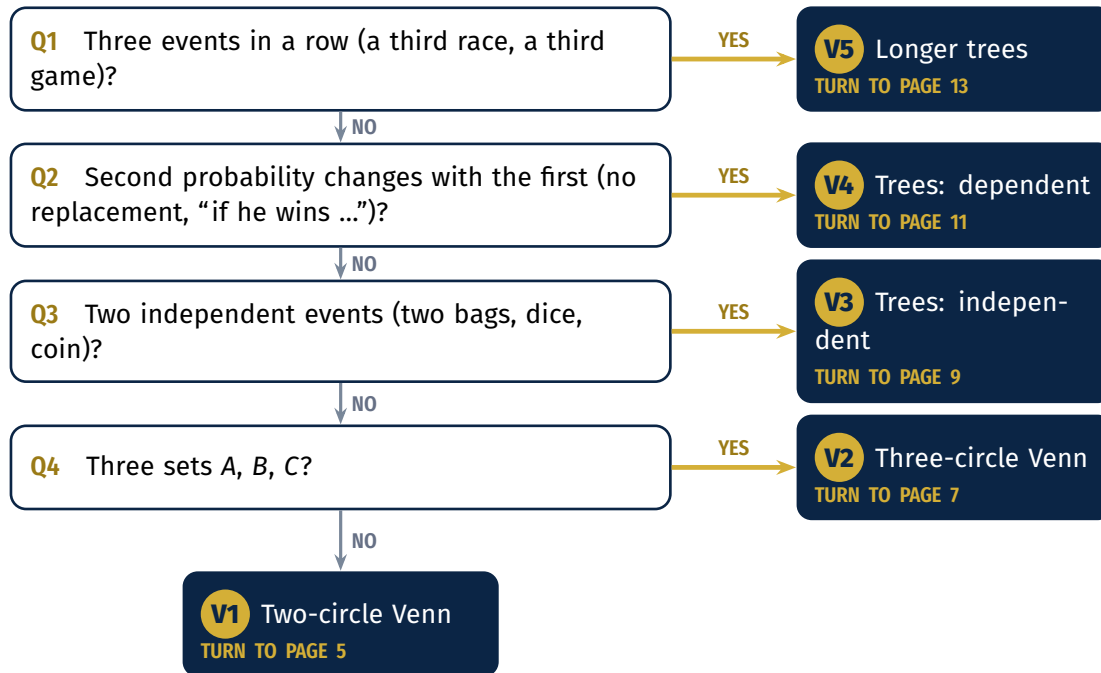
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Two-Circle Venn Diagrams

Overlap first: $n(A \cap B) = n(A) + n(B) - n(A \cup B)$;
then the rest.

SEEN IN PAST PAPERS

3 / 24

PRACTICE QUESTIONS: 18


Your notes

Came up in: Oct/Nov 2020 P1H Q1 • Jan 2021 P2HR Q16 • Nov 2025 P2H Q18

? What is it?

In plain words

Four regions: A only, both, B only, outside. Fill the **overlap first**, then subtract to get “only” regions, then the outside.

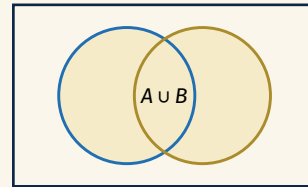
Probability = $\frac{\text{region}}{\text{total}}$.

IT LOOKS LIKE

$$n(A) = 62$$

$$n(A \cup B) = 68$$

$$A \cup B'$$



How do I do it?

1 Overlap

Given, or $n(A) + n(B) - n(A \cup B)$.

2 Only

$n(A)$ - overlap; $n(B)$ - overlap.

3 Outside

Total - everything inside.

4 Read

The regions asked; $\div$ total.

MIDDLE FIRST Then the “only” parts.

V1 Two-Circle Venn Diagrams

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

" $n(A) = 62, n(B) = 30, n(A \cup B) = 68$ " → **Overlap** = $62 + 30 - 68$

" $A \cup B$ " → **In A, or outside B**

"in set A but not in set B" → **A only**

**WORKED EXAMPLE****PAST PAPER**

January 2021 · Paper 2HR · Q16 · 5 marks

$n(A) = 62, n(B) = 30, n(A \cup B) = 68, n(E) = 80$. (a) Complete the Venn diagram. (b) Find (i) $P(A \cap B)$ (ii) $P(A \cup B')$.

STEP 1 $62 + 30 - 68 = 24$ ◀ overlap

STEP 2 A only 38, B only 6, outside 12 ◀ the rest

STEP 3 $P(A \cap B) = \frac{24}{80} = \frac{3}{10}$ ◀ (b)(i)

STEP 4 $A \cup B' = 38 + 24 + 12 = 74 : \frac{74}{80} = \frac{37}{40}$ ◀ (b)(ii)

ANSWER $\frac{3}{10}; \frac{37}{40}$

✓ $38 + 24 + 6 + 12 = 80$ ✓

**WATCH OUT**

× A only = 62

✓ $62 - 24 = 38$

$n(A)$ includes the overlap.

**EXAM TIP**

$A \cup B'$ is everything except "B only":
 $80 - 6$.

**Check yourself**

calculator first · answers on page 15

1.1

Oct/Nov 2020 P1H

Numbers 1–14; A only: 8, 10, 14. One number is picked. $P(\text{in A but not in B})$?

Answer:

1.2

Nov 2025 P2H

40 students: L only 5, $L \cap G$ x, G only 7, $G \cap S$ 2x, S only 4x, none 14. $P(L \text{ or } S)$?

Answer:

1.3

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$n(E) = 40, n(A) = 22,$
 $n(B) = 15, n(A \cap B) = 6$.
Find $P((A \cup B)')$.

Answer:

V2

VARIANT 2

Three-Circle Venn Diagrams

Centre first; then the three “two only” regions; then the “one only” regions; then outside.

SEEN IN PAST PAPERS

4 / 24

PRACTICE QUESTIONS: 4

 Your notes

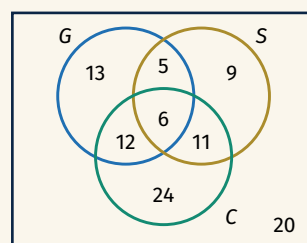
Came up in: Nov 2021 P1H Q16 • Jun 2022 P2H Q16 • Jun 2024 P2H Q16 • Jun 2025 P2H Q20

? What is it?

In plain words

Eight regions. “11 have sheep and goats” **includes** the 6 in all three — so only $11 - 6 = 5$ go in the sheep-and-goats-only region.

Work **inwards out**: centre, pairs, singles, outside. Unknown region? Use the total.



IT LOOKS LIKE

 $n(G' \cap C)$

given C

How do I do it?

1 Centre

All three.

2 Pairs

“A and B” – centre.

3 Singles

Circle total – the three parts inside it.

4 Outside

Given, or total – all inside.

CENTRE, PAIRS, SINGLES

Pairs include the centre.

V2 Three-Circle Venn Diagrams

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“11 have sheep and goats” → 11- centre in that region

“One farmer who has chickens ... also has goats” → $\frac{n(G \cap C)}{n(C)}$

WORKED EXAMPLE PAST PAPER

June 2022 · Paper 2H · Q16 · 8 marks

100 farmers: S 31, C 53, all 6, $S \cap G$ 11, $S \cap C$ 17, $G \cap C$ 18, none 20. Find $n(G)$ and $P(G | C)$.

STEP 1 centre 6; SG 5, SC 11, GC 12 ◀ pairs minus the centre

STEP 2 S only 9, C only 24 ◀ singles

STEP 3 G only = $100 - 20 - 67 = 13 \Rightarrow n(G) = 36$ ◀ use the total

STEP 4 $P = \frac{12+6}{53} = \frac{18}{53}$ ◀ given chickens

ANSWER $n(G) = 36; \frac{18}{53}$

✓ regions add to 100 ✓

WATCH OUT

✗ GC region = 18

✓ $18 - 6 = 12$

“Goats and chickens” includes the centre.

EXAM TIP

“Given that”: the denominator is that whole circle, not 100.

Check yourself

calculator first · answers on page 15

2.1 Nov 2021 P1H

M 25, $M \cap E$ 18, $M \cap F$ 11,
all three x, M only 3.
Find x; $P(\text{not } F | M)$.

Answer:

2.2 Jun 2024 P2H

All 9, $K \cap P$ 17, $P \cap E$ 16, P
28. $P(K | P)$?

Answer:

2.3 Jun 2025 P2H

C 43, all 12, $C \cap P$ 18,
 $C \cap T$ 27. Find $n(T' \cap C)$
and $P(P | C)$.

Answer:

V3

VARIANT 3 MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

11 / 24

PRACTICE QUESTIONS: 14

Tree Diagrams: Independent Events

Second branches are the same whatever happened first; multiply along, add between.

Came up in: Jan 2020 P2HR Q16 • Oct/Nov 2020 P1H Q15 • Jan 2022 P1HR Q13 • Jan 2022 P1H Q14 • Jun 2023 P2H Q13 • Nov 2023 P2H Q11 • May 2024 P1H Q13 • Jun 2025 P1H Q14 • Jun 2025 P2HR Q14 • Nov 2025 P1H Q13

 Your notes

? What is it?

In plain words

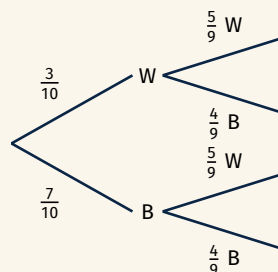
Two separate bags, two dice, a coin and a dice: the second pick **doesn't depend** on the first, so both second-stage pairs are **the same**.

Each pair adds to 1. "Same colour" = two routes; "at least one" = 1 - none.

IT LOOKS LIKE

bag A then bag B

"same colour"



How do I do it?

1 First stage

Each pair adds to 1.

2 Second stage

Same pair on every branch.

3 Routes

Multiply along each wanted route.

4 Add

Add the routes (or 1 - the opposite).

× **ALONG, + BETWEEN** The most common variant.

V3 Tree Diagrams: Independent Events

VARIANT 3 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"a bead from bag A and a bead from bag B"

→ **Independent: same second pair**

"two beads of the same colour"

→ **WW + BB**

"at least one ... green"

→ **1 - P(no green)****WORKED EXAMPLE****PAST PAPER**

May 2024 · Paper 1H · Q13 · 5 marks

Bag A: 3 white, 7 black. Bag B: 5 white, 4 black. One bead from each. Find the probability the beads are the same colour.

STEP 1 $A : \frac{3}{10}, \frac{7}{10}; B : \frac{5}{9}, \frac{4}{9}$ ◀ branches

STEP 2 $P(WW) = \frac{3}{10} \times \frac{5}{9} = \frac{15}{90}$ ◀ multiply along

STEP 3 $P(BB) = \frac{7}{10} \times \frac{4}{9} = \frac{28}{90}$ ◀ multiply along

STEP 4 $\frac{15}{90} + \frac{28}{90} = \frac{43}{90}$ ◀ add

ANSWER $\frac{43}{90}$

✓ less than 1, two routes ✓

**WATCH OUT**

✗ $\frac{3}{10} + \frac{5}{9}$
 ✓ $\frac{3}{10} \times \frac{5}{9}$

Along a route you multiply.

**EXAM TIP**

Fill in every branch before answering — the tree itself is worth marks.

**Check yourself**

calculator first · answers on page 15

3.1

Jun 2023 P2H

Dice A: 2 red, 4 yellow faces. Dice B: 1 red, 5 yellow. P(both yellow)?

Answer:

3.2

Oct/Nov 2020 P1H

Spinner A: 2 red, 3 green sections. Spinner B: 4 red, 1 green. P(at least one green)?

Answer:

3.3

Nov 2025 P1H

Bag: 2 yellow, 3 orange. Box: 3 yellow, 5 green. One from each. P(different colours)?

Answer:

V4

VARIANT 4

Tree Diagrams: Dependent Events

Second branches change: without replacement (one fewer) or “if ... then” probabilities.

SEEN IN PAST PAPERS

4 / 24

PRACTICE QUESTIONS: 8


Your notes

Came up in: Jan 2020 P2H Q14 • Jun 2022 P1H Q12 • May 2023 P1HR Q13 • Nov 2024 P1H Q13

? What is it?

In plain words

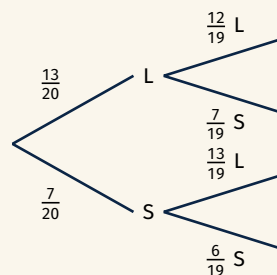
Without replacement: after taking one, the bag has **one fewer** — and one fewer of that colour.

“If he wins the first, the probability he wins the second is 0.9”: the **second pair depends** on the first branch.

IT LOOKS LIKE

two glasses taken

“if he uses the treadmill”



How do I do it?

1 First stage

As usual.

2 Second stage

Recount after each first outcome (or use the “if” values).

3 Routes

Multiply along.

4 Add

The wanted routes.

ONE FEWER ON THE BOTTOM

And on top if the same kind.

V4 Tree Diagrams: Dependent Events

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"takes at random two glasses"*→ **Without replacement: denominators 20, then 19***"If he wins the first game, the probability ... is 0.9"*→ **Different second pairs***"Complete the probability tree diagram"*→ **Each pair adds to 1****WORKED EXAMPLE PAST PAPER**

January 2020 · Paper 2H · Q14 · 4 marks

20 glasses: 13 large, 7 small. Roberto takes two at random. (a) Complete the tree. (b) Find $P(\text{two small glasses})$.

STEP 1 $\frac{13}{20}, \frac{7}{20}$ ◀ first glass

STEP 2 after L: $\frac{12}{19}, \frac{7}{19}$; after S: $\frac{13}{19}, \frac{6}{19}$ ◀ 19 left

STEP 3 $\frac{7}{20} \times \frac{6}{19}$ ◀ small then small

STEP 4 $= \frac{42}{380} = \frac{21}{190}$ ◀ simplify

ANSWER $\frac{21}{190}$

✓ all four routes add to 1 ✓

**WATCH OUT**

✗ $\frac{7}{20} \times \frac{7}{20}$
 ✓ $\frac{7}{20} \times \frac{6}{19}$

The first glass is not put back.

**EXAM TIP**

Write the numbers left in the bag next to each first-stage branch.

**Check yourself**

calculator first · answers on page 15

4.1

Jun 2022 P1H

$P(\text{treadmill}) = 0.8$; if treadmill, $P(\text{cross trainer}) = 0.3$. $P(\text{both})$?

Answer:

4.2

Jan 2021 P1H

Ding wins game 1 with 0.6; then wins game 2 with 0.9 (after a win) or 0.25 (after a loss). $P(\text{wins exactly one})$?

Answer:

4.3

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4 red, 6 blue beads; two taken without replacement. $P(\text{both blue})$?

Answer:

V5

VARIANT 5

Longer Trees: Three Stages

List every route that gives the result; multiply along each; add them.

SEEN IN PAST PAPERS

2 / 24

PRACTICE QUESTIONS: 4

 Your notes

Came up in: May 2021 P2H Q13 • Nov 2021 P1H Q15

? What is it?

In plain words

A third stage, or three outcomes per branch (win / draw / lose). Don't draw the whole tree — **list the routes** you need.

"Exactly two of three": WWL, WLW, LWW — with the right probability at each step.

$$\text{WWL: } 0.28 \times 0.4$$

$$\text{WLW: } 0.42 \times 0.3$$

$$\text{LWW: } 0.12 \times 0.3$$

LIST, MULTIPLY, ADD

IT LOOKS LIKE

third race

win/draw/lose

How do I do it?

1 Routes

Every order giving the result.

2 Each step

The right probability (it may depend on earlier results).

3 Multiply

Along each route.

4 Add

All the routes.

LIST THE ROUTES No need to draw 8 branches.

V5 Longer Trees: Three Stages

VARIANT 5 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“exactly two of the three races”

→ **Three routes**

“If she wins both ... third race is 0.6”

→ **Third-stage value depends on the first two**

“the same number of points”

→ **List the matching results****WORKED EXAMPLE****PAST PAPER**

May 2021 • Paper 2H • Q13(c) • 4 marks

Emilie wins race 1 with 0.7 and race 2 with 0.4. If she wins both, $P(\text{wins race 3}) = 0.6$; if exactly one, 0.3. Find $P(\text{wins exactly two of three})$.

STEP 1 $P(\text{both}) = 0.28$, $P(\text{exactly one}) = 0.7(0.6) + 0.3(0.4) = 0.54$ ◀ first two races

STEP 2 won both, lose 3rd : $0.28 \times 0.4 = 0.112$ ◀ route 1

STEP 3 won one, win 3rd : $0.54 \times 0.3 = 0.162$ ◀ routes 2 and 3

STEP 4 $0.112 + 0.162 = 0.274$ ◀ add

ANSWER 0.274

✓ less than 1 ✓

**WATCH OUT** $\times 0.7 \times 0.4 \times 0.6$

✓ that is three wins

Exactly two means one race is lost.

**EXAM TIP**

Group the routes by what happened in the first two stages.

**Check yourself**

calculator first • answers on page 15

5.1

Nov 2021 P1H

Each game: Magnus wins $\frac{2}{9}$, draw $\frac{4}{9}$. After 2 games, $P(\text{same points})$?

Answer:

5.2

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$P(\text{hit}) = 0.6$ for each of three shots, independently. $P(\text{at least one hit})$?

Answer:

5.3

Practice • Hard

$P(\text{pass first try}) = 0.7$; if not, $P(\text{pass the retake}) = 0.9$. $P(\text{pass})$?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1 $\frac{3}{14}$

1.2 $\frac{19}{40}$

1.3 $\frac{9}{40}$

2.1 $x = 7; \frac{14}{25}$

2.2 $\frac{17}{28}$

2.3 $16; \frac{18}{43}$

3.1 $\frac{5}{9}$

3.2 $\frac{17}{25}$

3.3 $\frac{17}{20}$

4.1 0.24

4.2 0.16

4.3 $\frac{1}{3}$

5.1 $\frac{28}{81}$

5.2 0.936

5.3 0.97

THE WHOLE TOPIC ON ONE PAGE

Venn & tree cheat sheet



Your notes
V1 Two circles 3/24

$$62 + 30 - 68 = 24$$

Overlap first.

V2 Three circles 4/24

$$11 - 6 = 5$$

Centre, pairs, singles.

V3 Independent 11/24

$$\frac{15}{90} + \frac{28}{90}$$

× along, + between.

V4 Dependent 4/24

$$\frac{7}{20} \times \frac{6}{19}$$

One fewer, or “if” values.

V5 Longer 2/24

$$0.112 + 0.162$$

List the routes.

**TOP 3 LOST MARKS**

1. pair regions include the centre × 2. adding along a route × 3. wrong total for “given that” ×

BOTTOM LINE Venn: middle first · tree: pairs add to 1 · × along, + between · given A: divide by $n(A)$.